



Figure 1: PandaBoard

- OMAP4 (Cortex-A9) CPU-based open development platform.
- OMAP4430 Application processor
- 1 GB low-power DDR2
- Display HDMI v1.3 Connector (Type A) to drive HD displays, DVI-D Connector
- 8.89 cm audio in/out and HDMI Audio out
- Full-size SD/MMC card
- Built-in 802.11 and Bluetooth v2.1+EDR
- On-board 10/100 Ethernet
- Expansion: 1xUSB OTG, 2xUSB HS host ports, general-purpose expansion header

PandaBoard is such a powerful mobile computing platform that one can port various OSs like Android and Ubuntu 10.04 to it!

Image-Processing (IP) software

There are many software and libraries available for image processing, like MATLAB, OpenCV, etc. Licensing costs for MATLAB are very high, but OpenCV is preferable since:

- It is free.
- It is fast.
- Has good documentation, tutorials, user groups, forums, etc.
- There are a lot of pre-built functions and algorithms to get a head start.
- There is active development on interfaces for other languages like Ruby, Python, MATLAB, etc.

OpenCV grew out of an Intel Research initiative to advance CPU-intensive applications. The intent behind OpenCV was to provide a platform that a student could readily use for developing applications, instead of reinventing basic functions from scratch. Figure 2 is a block diagram showing the different modules in OpenCV.

Arduino

To make any changes in the external environment, we need a hardware circuit with a 'brain'. This role can be filled by any microcontroller or microprocessor. Various companies such as Texas Instruments, NXP, Maxim, Atmel, etc, make microcontrollers. A microcontroller cannot be directly used in a circuit, as is. A full development circuit board has to be made, providing access to different pins of the microcontroller. This process is not easy, as it involves designing the circuit, soldering, etc. All changes are permanent. Also, a program

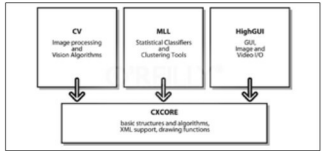


Figure 2: OpenCV modules

loader has to be made to load the code from the computer into the microcontroller. This can be quite cumbersome, so Arduino is used, due to the following reasons:

- It is open source, so all schematics are available and you can design the same board on your own.
- It is based on the concept of breadboard prototyping, which encourages the use of hook-up wires and breadboard, to make a circuit which can be easily modified later. This avoids the hassles of soldering.
- There are a number of tutorials, tons of code, forums, etc, to help beginners.
- The design of the board and IDE is so simple that one does not have to be an engineer to use it. Even a 6th grade student can use it.
- It is pretty cheap compared to other available development boards.

There are various versions of Arduino to choose from, such as Uno, Mega, Lilypad, etc. I personally prefer the Arduino Mega (Figure 3), which is a microcontroller board based on the ATmega1280. It has 54 digital input/output pins, of which 14 can be used as PWM (Pulse-width modulation) outputs, 16 analogue inputs, 4 UARTs (hardware serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable, or power it with an AC-to-DC adaptor or battery to get started.

Pyserial

This is a library that provides Python support for serial connections (RS-232) over a variety of different devices: old-style serial ports, Bluetooth dongles, infra-red ports, and so on. It also supports remote serial ports via RFC 2217 (since V2.5). We use this for communication with the external hardware (Arduino). This library is easy to use, and has good documentation.

Python

Python is a general-purpose, high-level programming language whose design philosophy emphasises code readability. Python claims to combine remarkable power with very clear syntax, and its standard library is large and comprehensive.

OpenCV installation

Let us now proceed to OpenCV installation. All commands are meant to be run in a terminal.